CTU TRAINING SOLUTIONS

Diploma in IT in Network Design and Administration

NET731 - Week 5 Report

Role-Based Access Control, NSG, and Storage Configuration

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Module	NET731
Assessment	Friday Practical Assessment (FA3)
Week	Week 5 - RBAC and Security Configuration

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Introduction

The purpose of Week 5 was to configure access control, network security, and storage services in Microsoft Azure. The practical work focused on assigning role-based access control roles, configuring inbound and outbound Network Security Group rules, creating a storage account, creating a blob container, uploading a sample file, and assigning storage access permissions.

Week 5 Task Overview

Week 5 requires students to assign RBAC roles to users in the resource group, configure NSG rules for inbound and outbound traffic, create an Azure Storage Account, configure containers and upload sample files, and set user access permissions for blob or file access.

RBAC Configuration at Resource Group Scope

The Reader role was assigned at the scope of the resource group CTU-20231149-RG by using Access control (IAM). The existing student account, Phenyo Tshifhenye, was selected as the member for testing the role assignment. Using the Reader role demonstrates basic RBAC access because it allows visibility of resources without allowing configuration changes. This step confirmed that RBAC can be applied successfully at resource group level in the Azure environment.

Network Security Group Configuration

The existing Network Security Group CTU-NET731-NSG was updated with additional security rules. An inbound rule named Allow-SSH-Inbound was created to allow TCP port 22 traffic with priority 300. This supports secure remote administration of the virtual machine over SSH. An outbound rule named Allow-HTTPS-Outbound was also created to allow TCP port 443 traffic with priority 300. This supports secure outbound communication over HTTPS when Azure resources need to access web services or updates.

Azure Storage Account Configuration

A storage account named phenyostorage18 was created inside the resource group CTU-20231149-RG in the South Africa North region. The account was configured with the Standard performance tier and Locally-redundant storage replication.

Container Configuration and File Upload

Inside the storage account, a blob container named week5files was created. A sample file was then uploaded successfully into the container. It demonstrated that Azure Blob Storage was working correctly and that the storage account could be used to organise and store files in the cloud for later access and management.

Storage Access Permission

To control access at storage account level, the Storage Blob Data Contributor role was assigned to Phenyo Tshifhenye at the scope of phenyostorage18. This role provides permission to read, write, and delete blob data.

Screenshot Evidence

Figure 1. RBAC role assignment opened at resource group scope

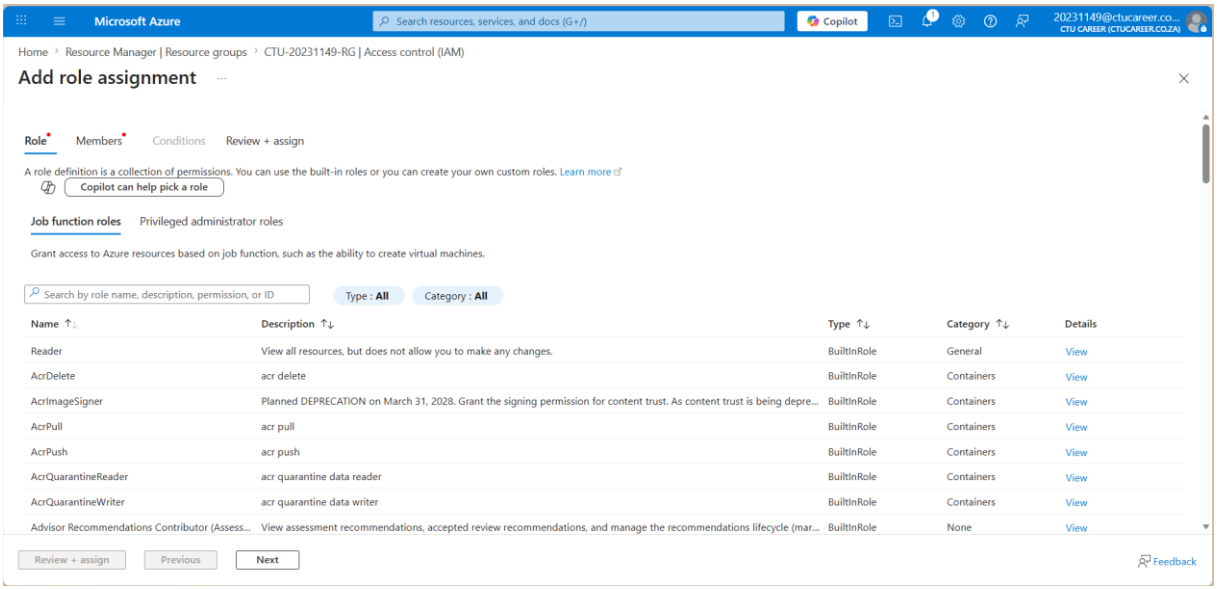


Figure 2. RBAC Reader role review and assignment at CTU-20231149-RG

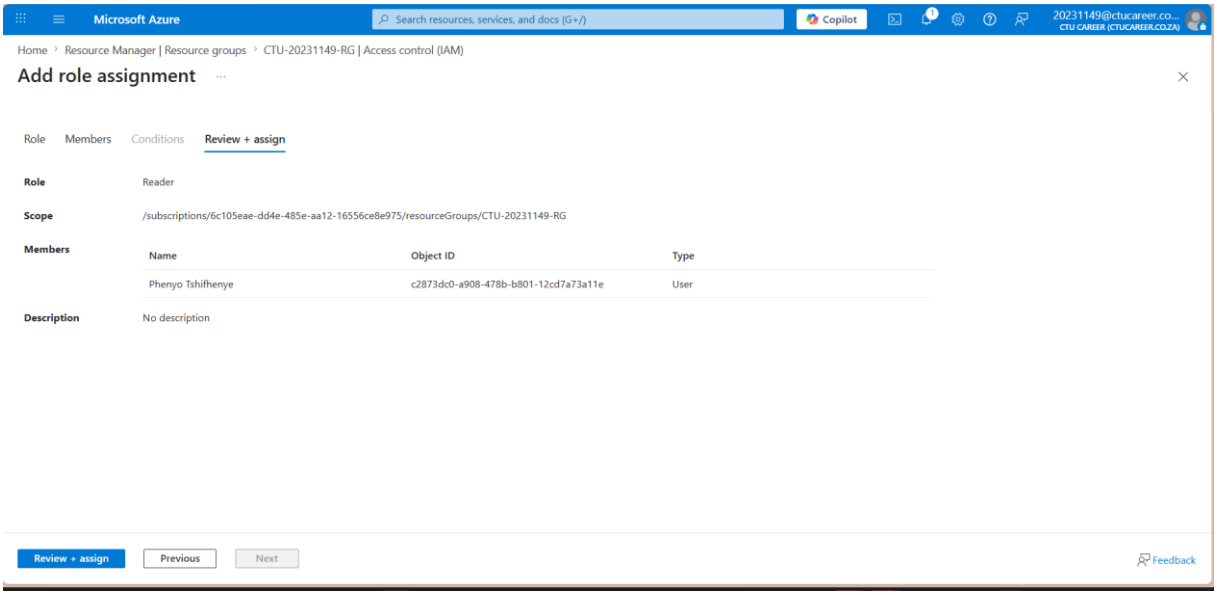


Figure 3. Reader role shown in the resource group role assignment list

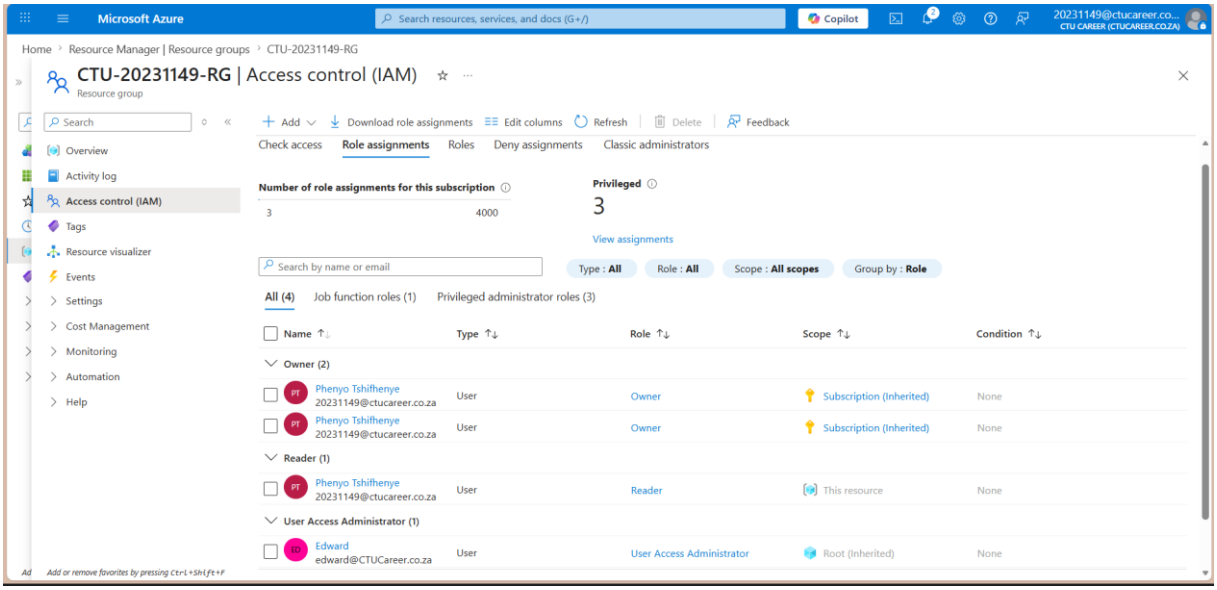


Figure 4. Inbound SSH rule configured before saving

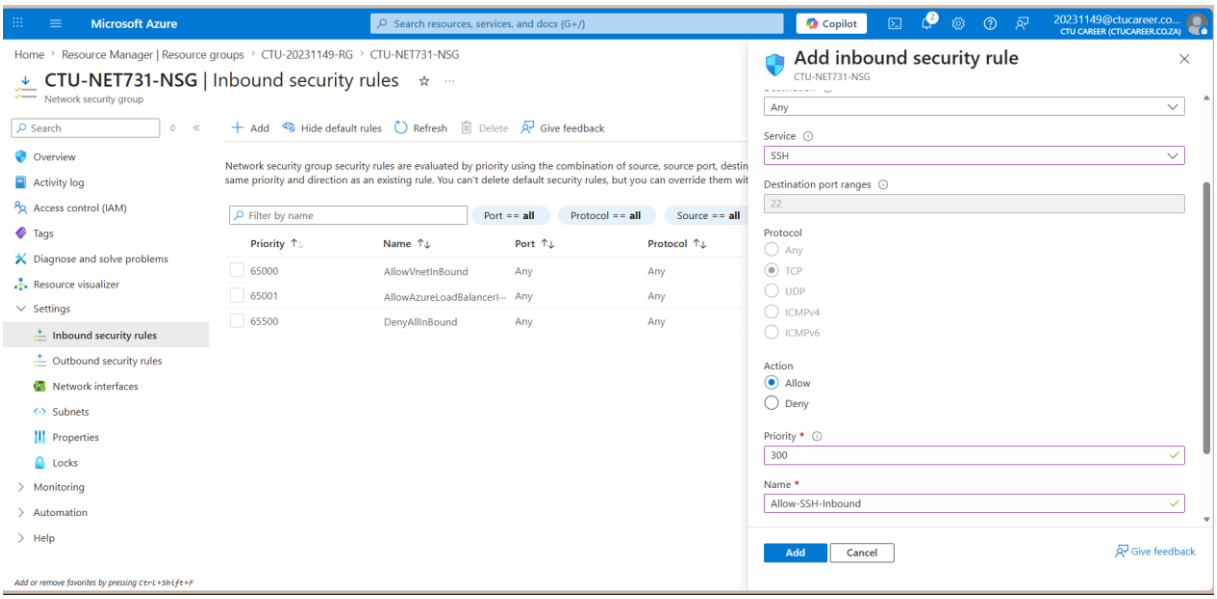


Figure 5. Inbound SSH rule shown in the NSG rule list

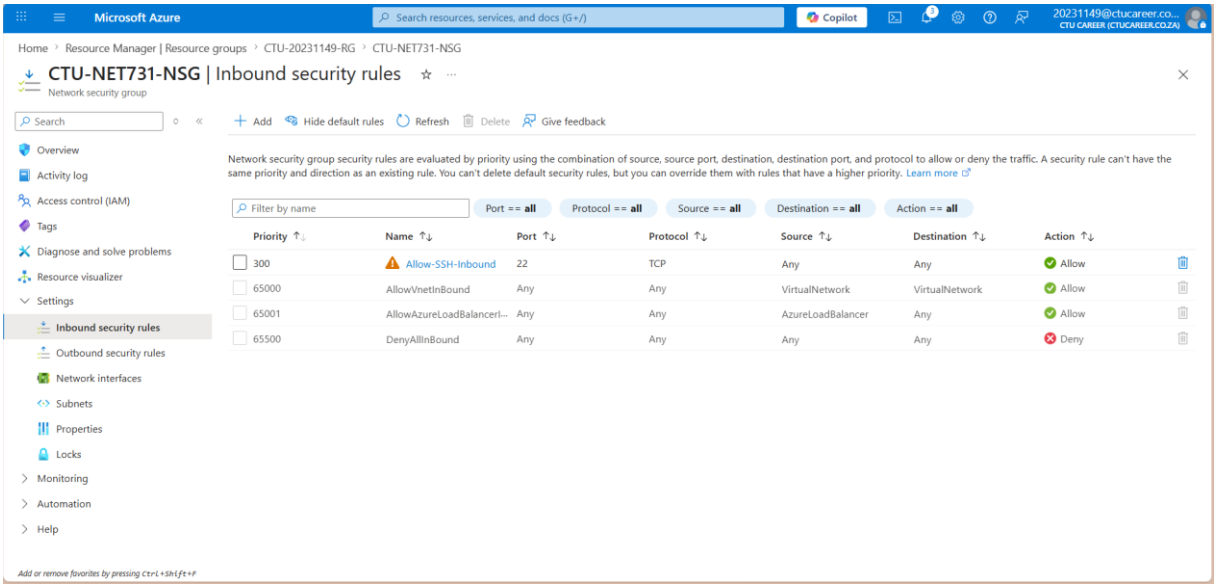


Figure 6. Outbound HTTPS rule configured before saving

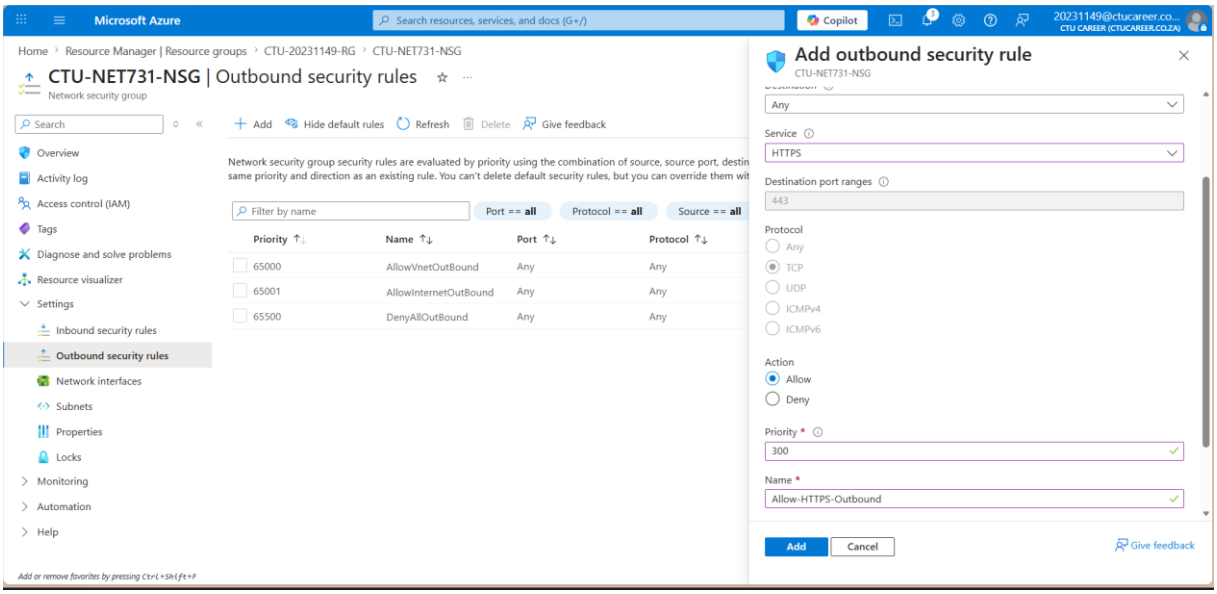


Figure 7. Outbound HTTPS rule shown in the NSG rule list

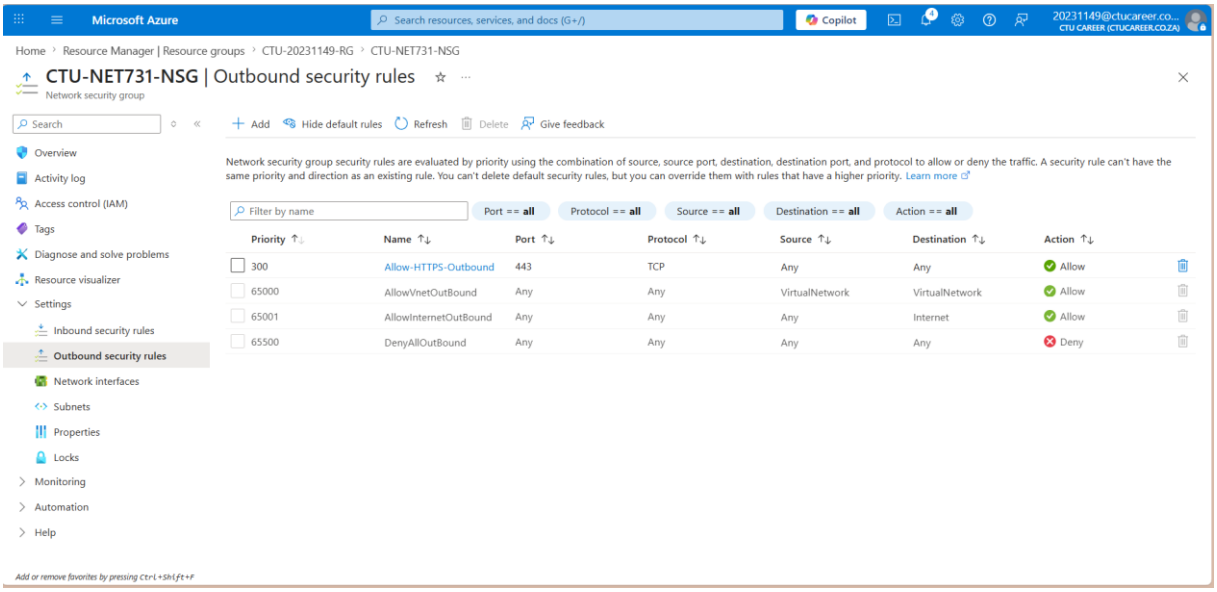


Figure 8. Storage account basics configuration

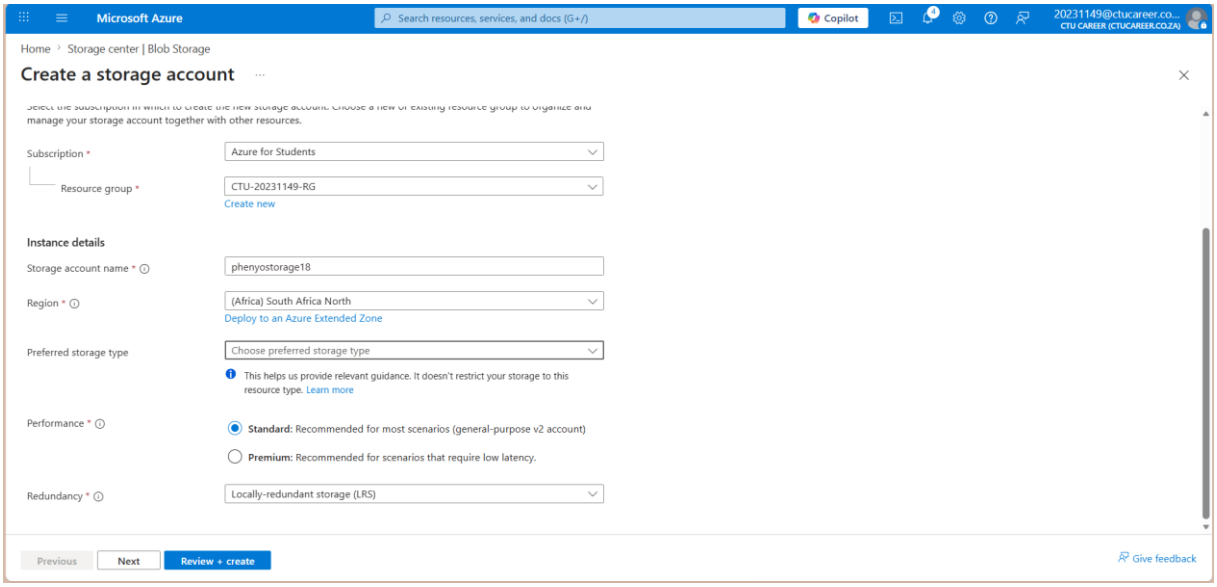


Figure 9. Storage account overview after deployment

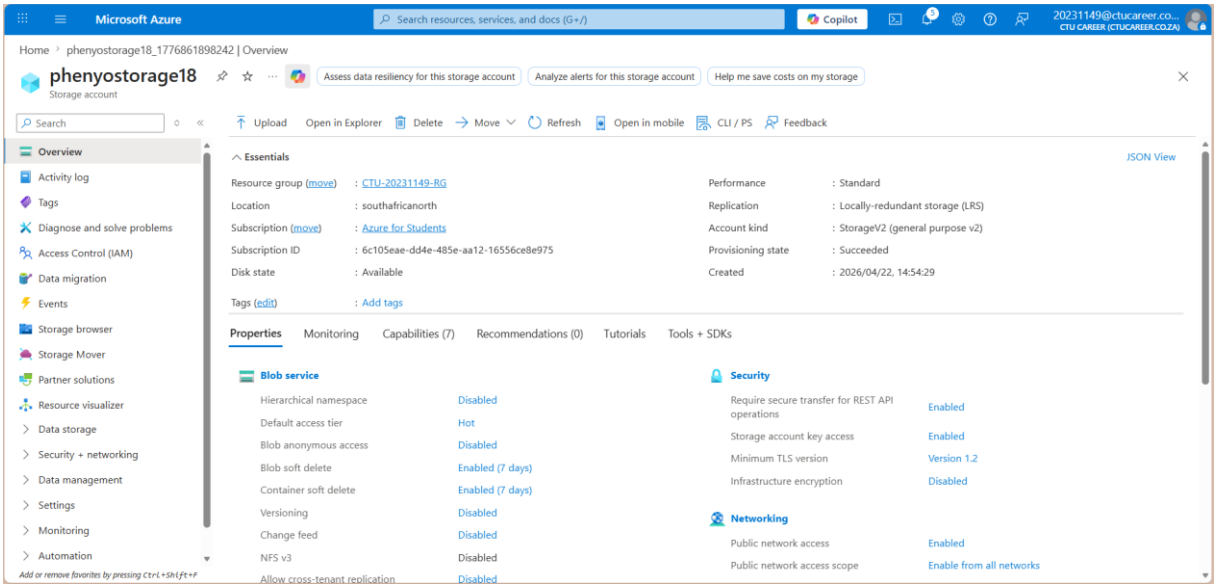


Figure 10. week5files container shown in the container list

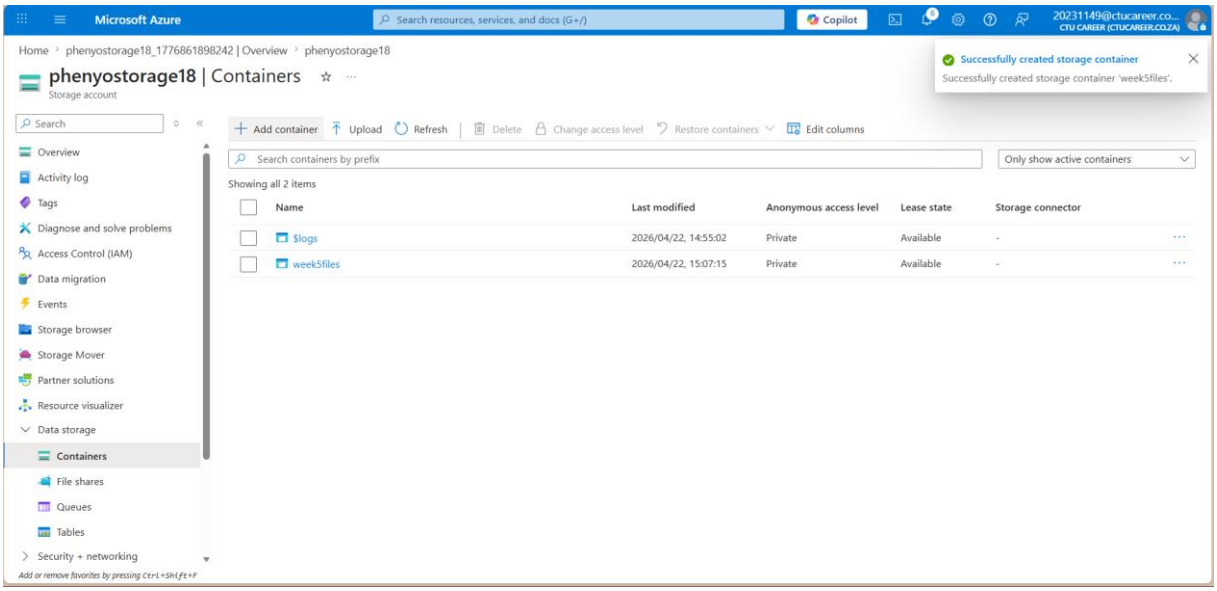


Figure 11. Sample file uploaded successfully to the week5files container

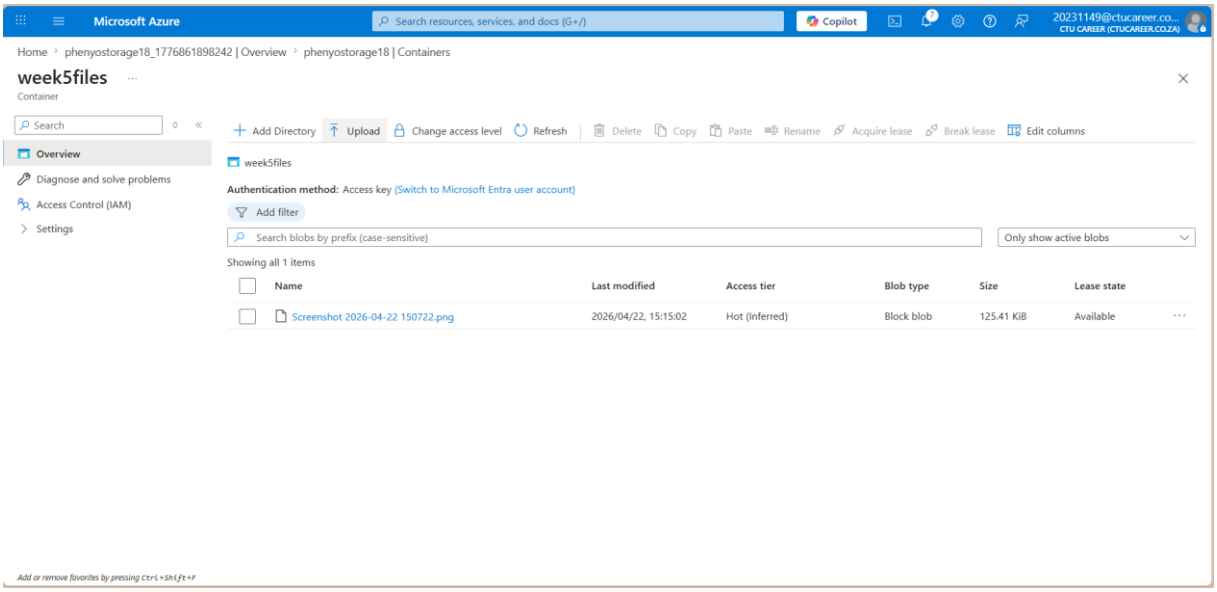


Figure 12. Storage Blob Data Contributor selected at storage account scope

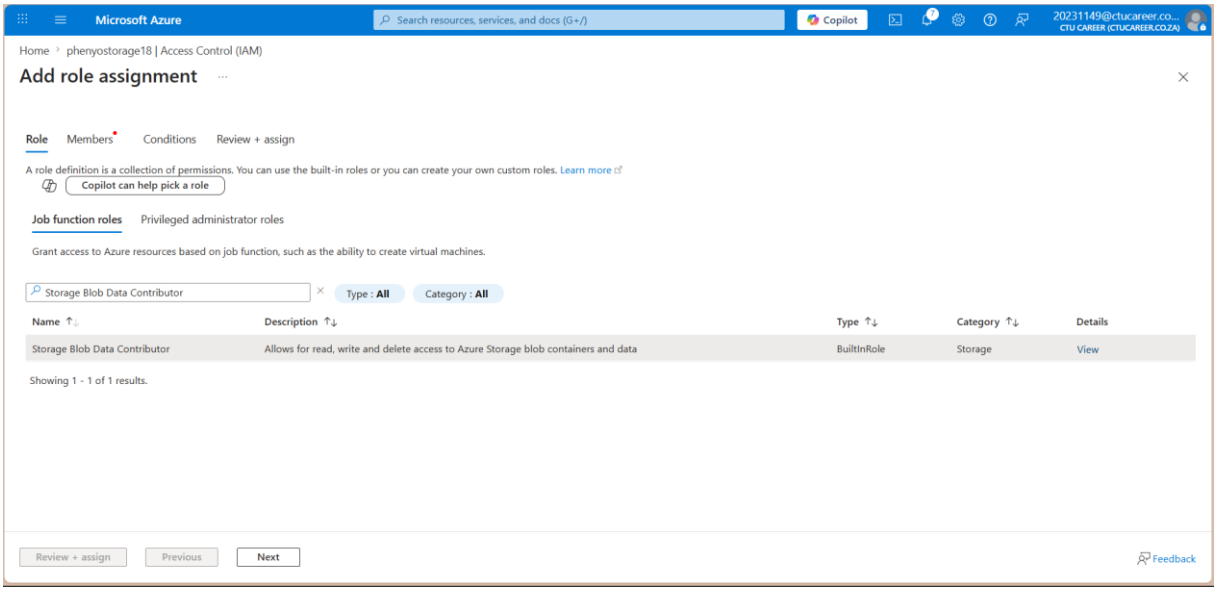


Figure 13. Storage Blob Data Contributor review and assignment on phenyostorage18

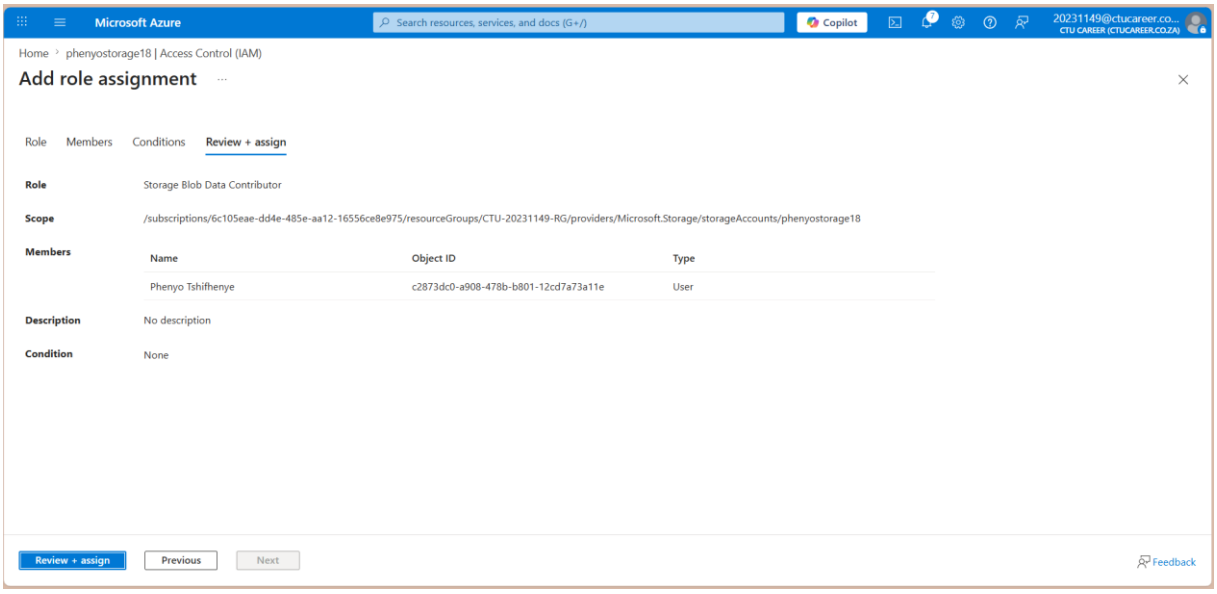
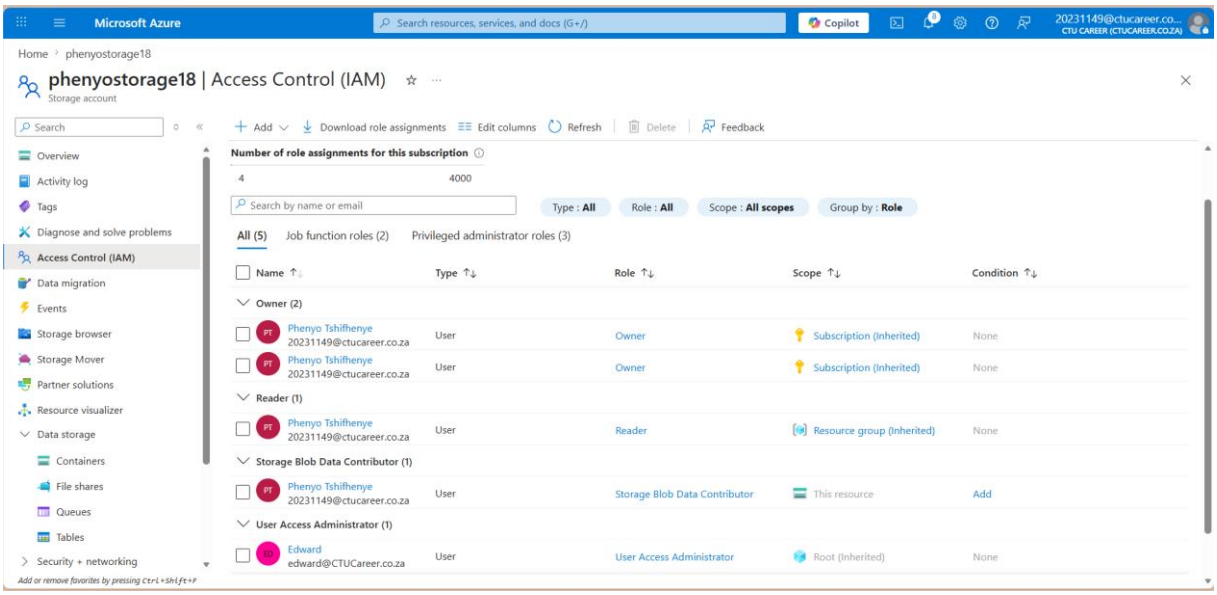


Figure 14. Storage Blob Data Contributor shown in the storage account role list



Conclusion

Week 5 was completed successfully by combining RBAC, NSG rules, and Azure Storage configuration. The completed work shows that access can be controlled at both resource group and storage account scope, network traffic can be filtered with NSG rules, and files can be stored and managed through Azure Blob Storage.

References

1. Microsoft (2026) Assign Azure roles using the Azure portal - Azure RBAC. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/azure/role-based-access-control/role-assignments-portal
2. Microsoft (2026) Create, change, or delete Azure Network Security Groups. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/azure/virtual-network/manage-network-security-group
3. Microsoft (2026) Create an Azure storage account. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/azure/storage/common/storage-account-create
4. Microsoft (2026) Quickstart: Upload, download, and list blobs - Azure portal. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/azure/storage/blobs/storage-quickstart-blobs-portal